Burning Up

With prices falling and media becoming more readily available, almost everyone can afford a dual-layer DVD writer. But which one to buy? That's where we come in...

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It's that time of the year again, and we set out in search for all the optical drives we could get our hands on. Surprisingly, variety doesn't seem to be sought after when it comes to DVD writers, as we only found eight drives to be the popular choice amongst retailers. In addition, we also got our hands on a couple of Blu-ray drives to see what the near future has in store for us.

The Drives

We got DVD Writers from different manufacturers. Some drives came from the old warhorses such as ASUS, Samsung and Sony, and some from newer entrants such as Moser Baer and Philips. In fact, Moser Baer makes its DVD Writers in collaboration with Lite-On—a famous brand of DVD Writers. Philips DVD Writers have been around abroad for many years, and are now available across India.

Interestingly, most optical drives are now SATA drives, and IDE seems to be dying of old age. The drives we got with the SATA interface were ASUS DRW-1814BLT, ASUS DRW-2014L1T, Philips SPD2514T, Samsung SH-S203, and Sony DRU-845S, while those with the IDE interface were Moser Baer DH-20A4P23C, Philips SPD2414T, and Sony DRU-190A.

Chaitanya Surpur

FEATURES

Supported Speeds

In our last test, we saw one DVD writer dominate the rest with a maximum write speed for DVD-Rs of a whopping 20x. This time around, almost all the drives support 20x writes, with none going faster, because we seem to have reached the maximum speeds for DVDs. Only the ASUS DRW-1814BLT supported a maximum write speed of 18x, which shoul still be fast enough for most. What's funny here is that its reading speed tops out at 16x—giving us a DVD writer that can write faster than it reads!

When it comes to DVD-RW and DVD+RW speeds, all the drives support 8x and 6x respectively. All the speed changes seem to have occurred in the dual-layer race, and although all the drives can write to DVD+R DL media at an acceptable 8x, the Sony DRU-845S is capable of 12x and the Samsung SH-S203 manages a whopping 16x! That's the fastest we've seen for dual-layer burning, ever.

With DVD+R DL media available for as little as Rs 100, and prices only seeming to fall more, there's never been a better time to invest in a dual-layer burner.

All the drives also feature the ability to write to DVD-RAM at 12x. The exception was the ASUS, which claims 14X DVD-RAM write speeds. DVD-RAM is great for video professionals, because you can treat your DVD-RAM drive as a hard drive—make folders, create files, even edit them from within the disc, etc. Availability of media, however, is a rather large thorn in the format's side.

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Additional Features

The two Philips drives are the only ones to not allow setting the BookType of DVD+R media when burning. Setting the BookType allows you to set the DVD you create to appear as a stamped DVD-ROM to standalone DVD players. This is helpful if your DVD player acts finicky when playing DVD-Video discs that are burnt on PCs.

Just like DVD-RAM, LightScribe is a feature that is fast losing ground due to waning interests. Again, expensive and hard to find media is the cause for this. LightScribe basically allows you to burn a CD/DVD label on to the disc, using the same drive that you're using to burn data on to it. You can create your own labels using text, images or both, and this custom label is burnt to the DVD/CD face using an additional laser. Only the ASUS drives—the DRW-1814BLT and DRW-2014L1T—support LightScribe this time.

Build Quality

When we talk about the build quality of an optical drive, it is the tray that we are talking about. We have no complaints about any of the trays this time, unlike a couple of years ago, when this was a serious issue with quite a few drives. The ASUS, Samsung and the Sony DRU-845S seem to have the best trays and load/eject mechanisms. Another thing to note here is that the ASUS, Moser Baer, and Samsung drives were

How We Tested

The drives were connected to the second SATA port or set as master and connected to the primary IDE channel, according to the type of the drive. The test process consisted of two sections: writing to and reading from different types of media.

Test Rig

Processor	Intel Pentium 4 540 3.2 GHz
RAM	1 GB of 667 MHz DDR2
Motherboard	ECS PF5 Extreme
Graphics Card	NVIDIA GeForce 7300GS
Hard Drive	Western Digital 80 GB SATA-II
Power Supply	Antec Neo480 480 Watt
OS	Windows XP Professional SP2
Burning software	Nero 8 Ultra Edition

The Reading Tests

Nero CD-DVD Speed was used to note the data transfer rates, access/seek times, CPU utilisation, burst rate, etc, for CDs as well as DVDs. This comes bundled with Nero 8 Ultra.

SiSoft Sandra Pro Business 2007, which is a system-wide performance benchmark as well as diagnostic utility, was used to measure the DVD read speeds and access times.

dbPowerAmp, an audio format conversion and CD ripping tool was used to rip three audio tracks of our test audio CD, located at different positions on the CD. The tracks were ripped to 44 KHz MP3 format at 128 Kbps, and the time was logged.

DVD Decrypter is a tool used to rip DVD video files to the hard drive. This was used to rip a select chapter (a .VOB file, 1 GB located towards the middle of the media) of our test DVD to the hard drive, and the time taken for this was noted.

The Writing Tests

These included logging the time taken to write to CD-R, CD-RW, DVD-R, DVD+R, DVD-RW and DVD+RW. An assorted data of 700 MB—the latest *Digit* CD—was used to write to CD-R, CD-RW and DVD-RW and the time was logged. Sequential data consisting of a 700 MB file—the same *Digit* CD zipped without compression—was used to write to a CD-R. Similarly assorted and sequential data of 4.38 GB was used to write to DVD-R and DVD+R media and the time taken for this was noted.

The Media

The media used were 52x CD-R from Moser Baer, 10x CD-RW from HP, 16x DVD-R from Sony, 16x DVD+R from Sony and 4x DVD-RW from HP. A point to note is that most of the DVD-Writers can burn DVDs at speeds in excess of 16x, but since such media is unavailable, the 16x media was burnt at the highest supported speed. We used Moser Baer DVD+R DL for dual layer media write test and an assorted data of 8.5 GB was burnt to these at their certified speeds of 8x where possible.

Other Features

Other features such as support for various DVD types, LightScribe capability, capability to set Book Type, tray build quality, silent operation and bundled accessories such as data cable, media, bezels, etc., were also noted and rated to derive the final score. Please note that the specified read and write speeds were not rated, instead we rated the read and write speeds in real world tests.